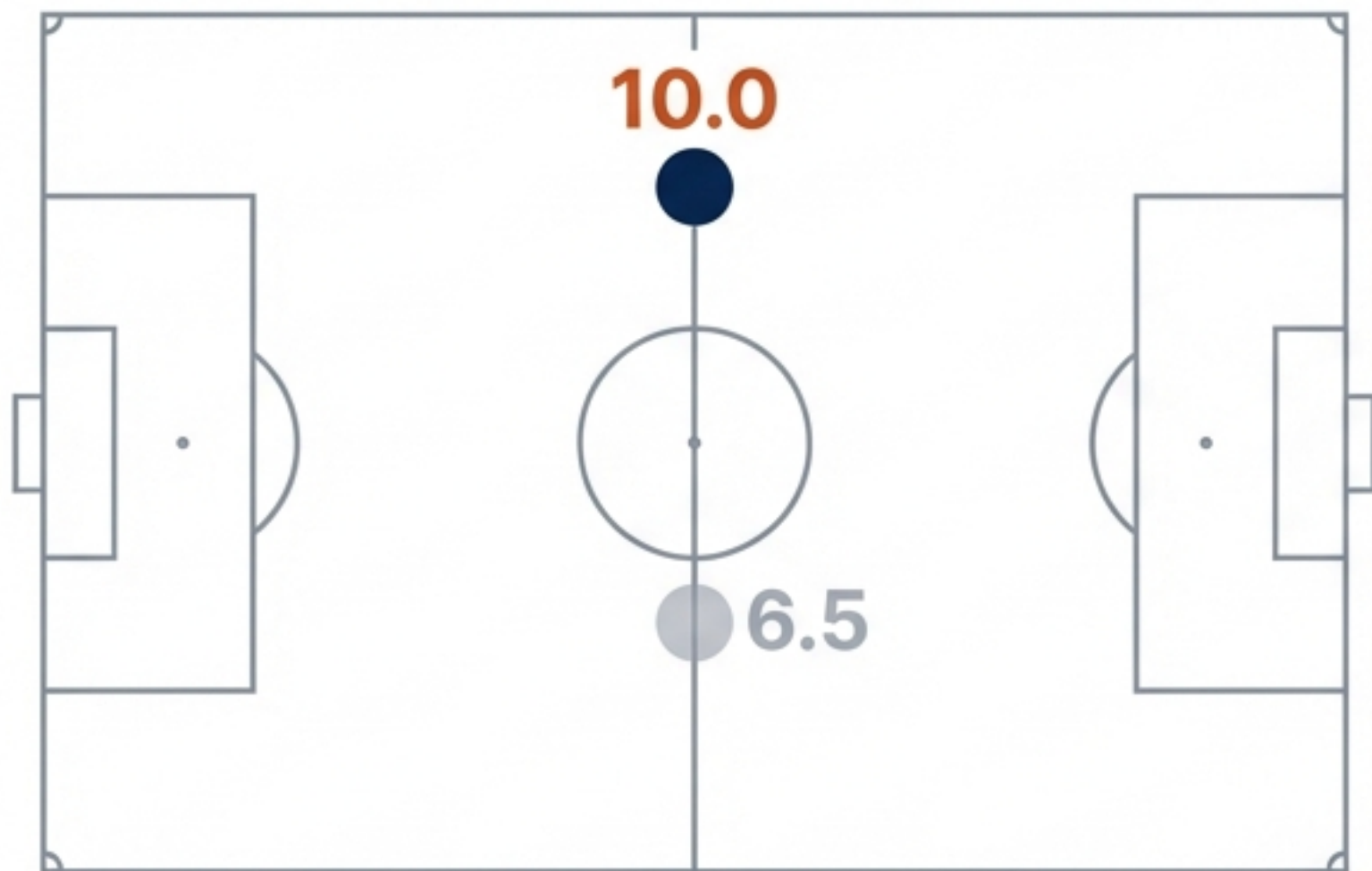


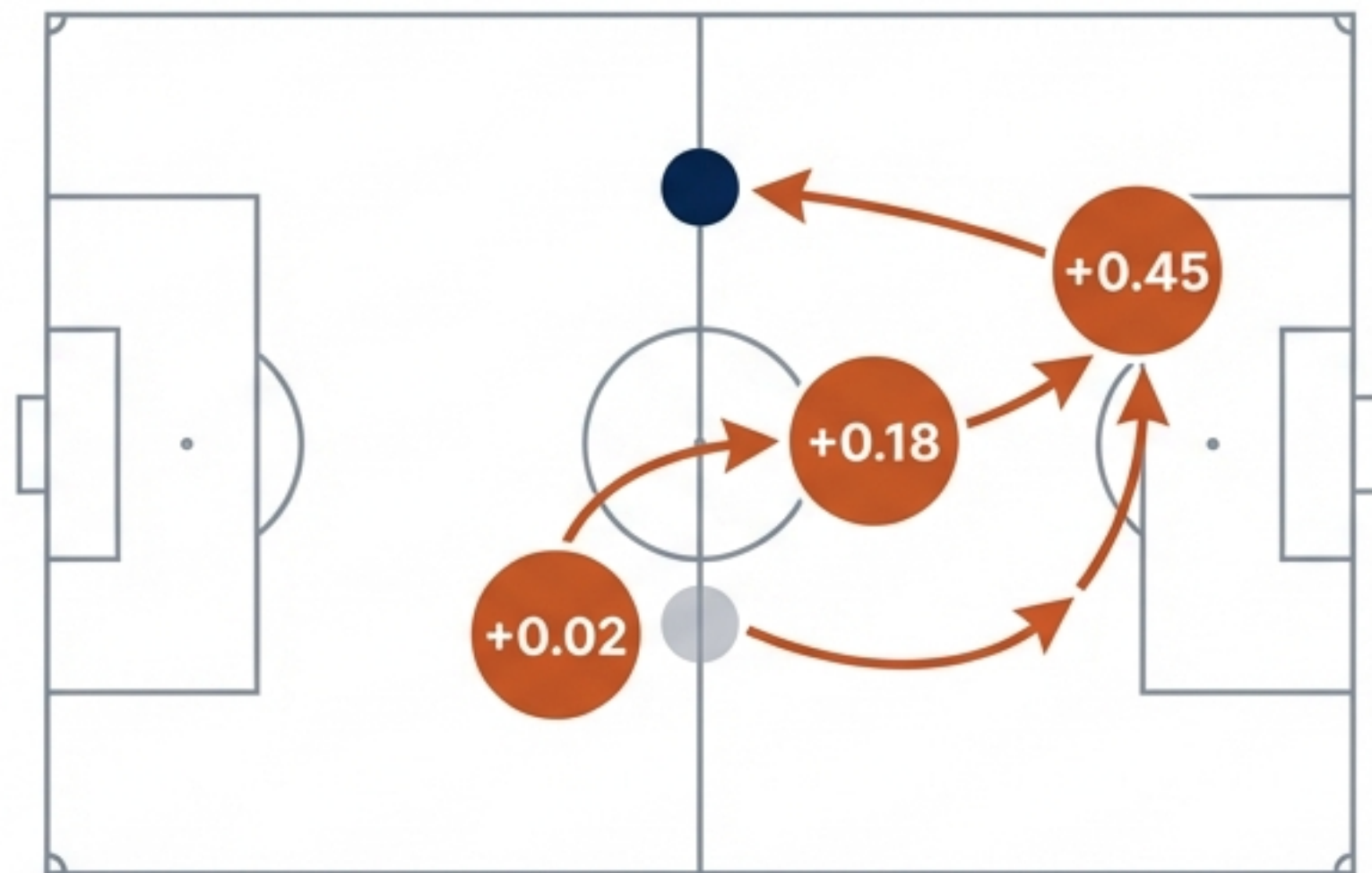
The FIFA Paradox: Overcoming Outcome Bias in Player Valuation

Outcome Bias: Rewards the destination, ignores the journey.



Traditional metrics (goals, assists, passing accuracy) disproportionately weight actions in the final third. They fail to capture the latent contributions that dictate match outcomes, systematically undervaluing tactical facilitators.

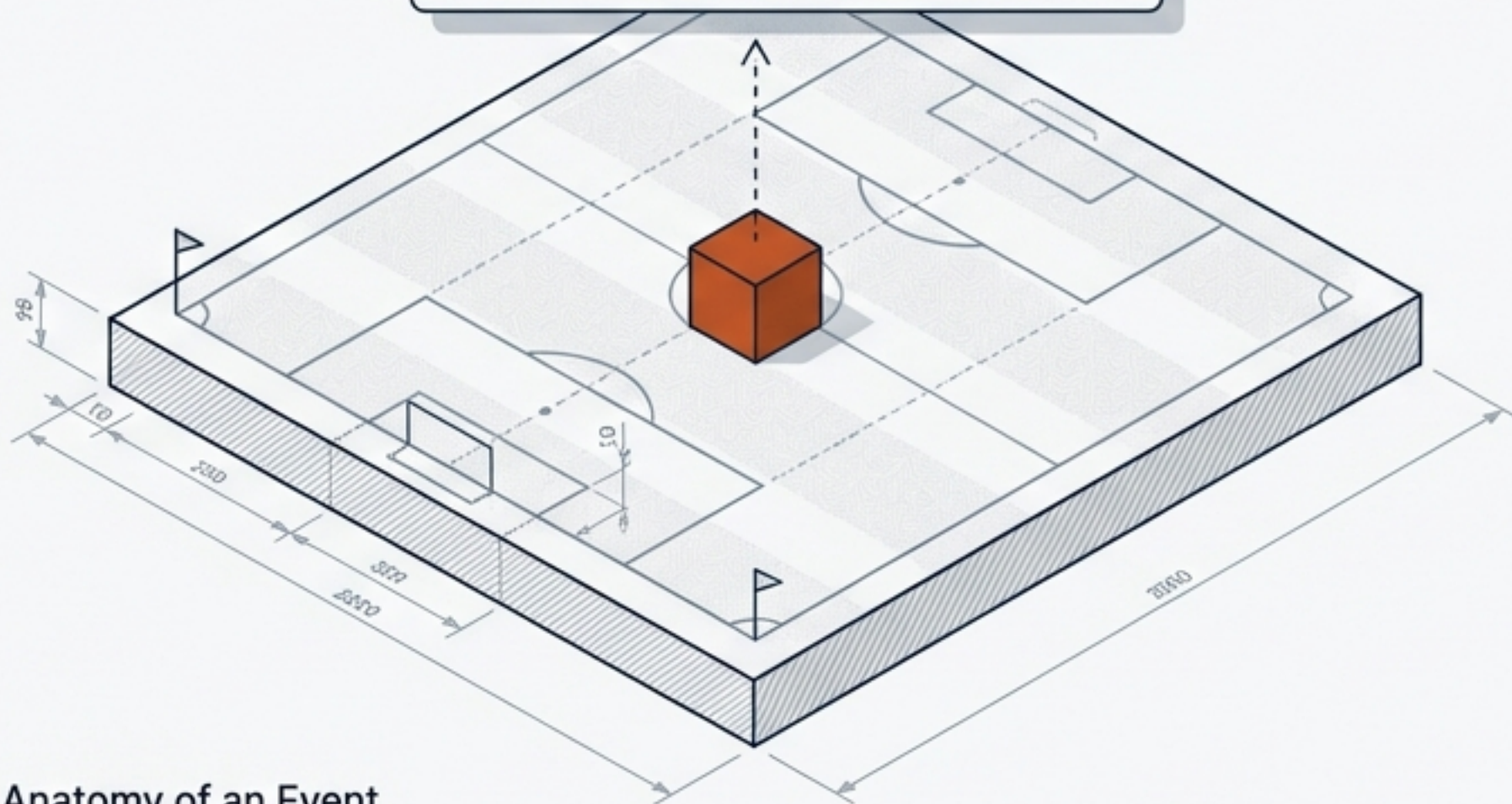
The Tactical Reality: Dominating State Transitions.



A single possession is a continuous chain of probability shifts. A deep-lying playmaker controlling the rhythm and manipulating the opposition's defensive block provides a massive structural advantage invisible to the box score.

High-Fidelity Foundation: StatsBomb Open Data

```
"type": "Pass"  
"location": [45.0, 60.0]  
"pass_end_location": [70.0, 85.0]  
"under_pressure": true  
"body_part": "Left Foot"  
"pass_height": "Ground Pass"
```



Anatomy of an Event

Beyond the Event Log

Moving past binary outcomes to capture 360-degree match context.

Spatial Precision

Industry-leading X/Y geolocation for every on-ball action (passes, dribbles, pressures, duels).

Contextual Depth

Capturing the exact circumstances of an action, including defensive pressure, body orientation, and precise pitch coordinates, providing the necessary mathematical substrate for complex state-value modeling.

The Engine: Constructing the Unified Performance Index (UPI)

Expected Goals (xG)

Evaluates the final execution. Treats shots as Bernoulli trials to estimate the isolated probability (0 to 1) of an attempt resulting in a goal based on location and angle.

Expected Threat (xT)



Evaluates spatial ball progression. A grid-based possession model quantifying territorial dominance by measuring the increase in threat when moving the ball from low-value to high-value pitch zones.

Unified Performance Index (UPI)

VAEP

Introduces the risk/reward trade-off. Accounts for defensive actions (tackles, interceptions) and the probability of conceding.

$$V(S_i) = P_{score}(S_i) - P_{concede}(S_i)$$

Sarah Rudd's Markov Chains



The underlying probabilistic framework. Links discrete game states across the entire possession chain, rewarding the multi-touch sequences that shift the state-value matrix before the final shot is ever taken.

Foundational Literature & Theoretical Precedents

